

Problem 22.1

$$\det(A - \lambda I) = (4 - \lambda)(2 - \lambda) = 0 \rightarrow \lambda = 4, 2$$

$\rightarrow S =$ All matrix with 4, 2 multiples on diagonal.
 Same for A^{-1}

Problem 22.2

$$\det(A - \lambda I) = (.6 - \lambda)(.1 - \lambda) - .36$$

$$= .06 - \frac{\lambda}{10} - \frac{6\lambda}{100} + \lambda^2 - \frac{36}{100}$$

$$= -\frac{7\lambda}{10} + \frac{\lambda^2}{100} - \frac{30}{100} = 0$$

$$-70\lambda + 100\lambda^2 - 30 = 0$$

$$100\lambda^2 - 70\lambda - 30 = 0$$

$$\lambda = 1, -0.3$$

$$\begin{bmatrix} -.4 & .9 \\ .4 & -.9 \end{bmatrix} X = 0 \rightarrow X = c \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} .9 & .9 \\ .4 & .4 \end{bmatrix} X = 0 \rightarrow X = c \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$S = \begin{bmatrix} 9 & 1 \\ 4 & -1 \end{bmatrix}$$

$$\wedge^k \xrightarrow{k \rightarrow \infty} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$S \wedge^k S^{-1} \xrightarrow{k \rightarrow \infty} \frac{1}{13} \begin{bmatrix} 9 & 9 \\ 4 & 4 \end{bmatrix}$$

the steady
state vector.